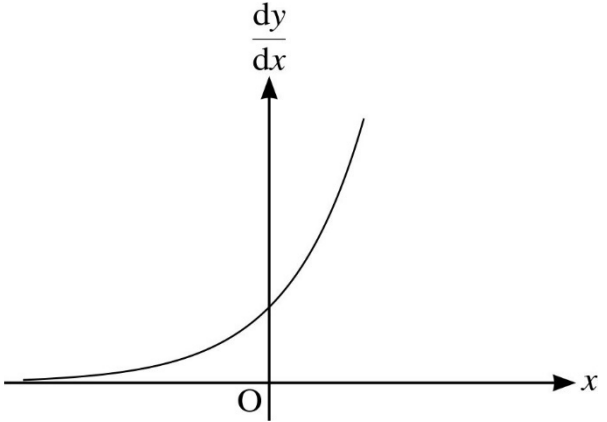
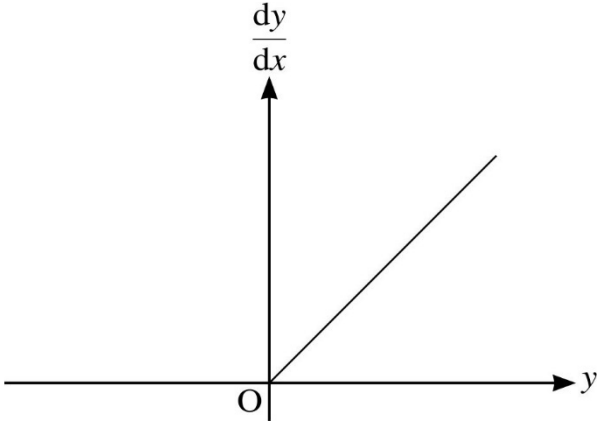
5	**(a)**	**(i)**		**B1**	**1.2**		

5	(a)	(ii)		M1 A1 [3]	2.2a 2.2a	Straight line (+ve gradient) through origin (may stop short of origin) First quadrant only	
5	(b)	(i)	Suitable reason e.g. • Reasonable to assume population growth is proportional to population • Populations are often modelled by exponential growth	E1 [1]	3.3	Allow e.g. wolves give birth to more wolves than they started with Do not allow e.g. population is proportional to time	
5	(b)	(ii)	$A = 21$ $51 = 21e^k$ $k = ln\left(\frac{51}{21}\right) = ln\left(\frac{17}{7}\right) \approx 0.887$ or better	B1 M1 A1 [3]	3.4 3.4 1.1	Allow 0.89	
5	(b)	(iii)	Suitable reason, e.g. • Population cannot keep growing • The wolves will run out of food if the population gets too big	E1 [1]	3.5b	Allow e.g. lack or resources or deforestation	